

Assignment No 11

Name: Sciddhanto Sinha

Class: CSE, SY-2(B)

Roll No.:2213111

Enrollment No.: MITU21BTCS0553

Aim:

Write a Python GUI program for the student registration and store the details in the file;
Allow the users to read the details from the file.

Theory:

Python provides the Tkinter toolkit to develop GUI applications.

Importing tkinter is same as importing any other module in the Python code. Note that the name of the module in Python 2.x is 'Tkinter' and in Python 3.x it is 'tkinter'.

```
import tkinter
```

There are two main methods used which the user needs to remember while creating the Python application with GUI.

Tk(screenName=None, baseName=None, className='Tk', useTk=1): To create a main window, tkinter offers a method 'Tk(screenName=None, baseName=None, className='Tk', useTk=1)'. To change the name of the window, you can change the className to the desired one. The basic code used to create the main window of the application is:

m=tkinter.Tk() where m is the name of the main window object

mainloop(): There is a method known by the name mainloop() is used when your application is ready to run. mainloop() is an infinite loop used to run the application, wait for an event to occur and process the event as long as the window is not closed.

```
m.mainloop()
```

```
import tkinter
```

```
m = tkinter.Tk()
```

```
'''
```

```
widgets are added here
```

```
'''
```

`m.mainloop()`

tkinter also offers access to the geometric configuration of the widgets which can organize the widgets in the parent windows. There are mainly three geometry manager classes class.

There are a number of widgets which you can put in your tkinter application. Some of the major widgets are explained below:

Button: To add a button in your application, this widget is used.

The general syntax is:

`w=Button(master, option=value)`

Label: It refers to the display box where you can put any text or image which can be updated any time as per the code.

The general syntax is:

`w=Label(master, option=value)`

master is the parameter used to represent the parent window.

Listbox: It offers a list to the user from which the user can accept any number of options.

The general syntax is:

`w = Listbox(master, option=value)`

master is the parameter used to represent the parent window.

MenuButton: It is a part of top-down menu which stays on the window all the time. Every menubutton has its own functionality. The general syntax is:

`w = MenuButton(master, option=value)`

master is the parameter used to represent the parent window.

Message: It refers to the multi-line and non-editable text. It works same as that of Label.

The general syntax is:

`w = Message(master, option=value)`

master is the parameter used to represent the parent window.

RadioButton: It is used to offer multi-choice option to the user. It offers several options to the user and the user has to choose one option.

The general syntax is:

`w = RadioButton(master, option=value)`

Scale: It is used to provide a graphical slider that allows to select any value from that scale.

The general syntax is:

w = Scale(master, option=value)

master is the parameter used to represent the parent window.

Scrollbar: It refers to the slide controller which will be used to implement listed widgets.

The general syntax is:

w = Scrollbar(master, option=value)

master is the parameter used to represent the parent window.

Text: To edit a multi-line text and format the way it has to be displayed.

The general syntax is:

w =Text(master, option=value)

Program:

```

from tkinter import *
from tkinter import ttk

def registration():
    name1 = name.get()
    contact1= contact.get()
    email1 = email.get()
    gender1 = gender.get()
    city1 = city.get()
    state1 = state.get()

    if name1== " or contact1==" or email1 == " or gender1 ==" or city1 ==" or state1 == ":
        message.set("Fill all the Details ")

    else:
        with open("Studinfo.txt", "a") as fp:
            if gender==1:
                fp.write(" "+name1+ " "+contact1+" "+email1+ " Male "+ city1+" "+state1 )
            else:
                fp.write(" " + name1 + " " + contact1 + " " + email1 + " Female " + city1 + " " +
state1)
            message.set("Data Stored Successfully")

def registrationform():

    global registration_screen

    registration_screen = Tk()

```

```
registration_screen.title("Registration form ")
```

```
registration_screen.geometry("400x400")
```

```
global message
```

```
global name
```

```
global contact
```

```
global email
```

```
global gender
```

```
global city
```

```
global state
```

```
name = StringVar()
```

```
contact = StringVar()
```

```
email = StringVar()
```

```
gender = IntVar()
```

```
city = StringVar()
```

```
state = StringVar()
```

```
message = StringVar()
```

```
Label(registration_screen, width="250", text="Please Enter the details below",bg='orange',  
fg='white').pack()
```

```
Label(registration_screen, text="Name ").place(x=25, y = 40)
```

```
Entry(registration_screen, textvariable=name).place(x=85, y = 41)
```

```
Label(registration_screen, text="contact ").place(x=25, y=80)
```

```
Entry(registration_screen, textvariable=contact).place(x=85, y=81)
```

```
Label(registration_screen, text="email ").place(x=25, y=120)
```

```
Entry(registration_screen, textvariable=email).place(x=85, y=121)
```

```
Label(registration_screen, text="Gender ").place(x=25, y=160)
```

```
Radiobutton(registration_screen, text= 'Male', variable=gender, value=1).place(x = 85, y =  
161 )
```

```
Radiobutton(registration_screen, text= 'Female', variable=gender, value=2).place(x = 150,  
y = 161 )
```

```
Label(registration_screen, text="City ").place(x=25, y=200)
```

```
citychoosen = ttk.Combobox(registration_screen, width= 30, textvariable=city)
```

```
citychoosen['values'] = ('Mumbai', 'Pune', "Chennai", "Kolkata", "Bengluru", "Patna",  
"Nagpur" )
```

```
citychoosen.current()
```

```
citychoosen.place(x = 85, y = 201)
```

```
Label(registration_screen, text="State ").place(x=25, y=240)
```

```
statechoosen = ttk.Combobox(registration_screen, width=30, textvariable=state)
```

```
statechoosen['values'] = ('Maharastra', 'Karnataka', "Andra Pradesh", "West Bangal",  
"Bihar", "Meghalaya", "Uttar Pradesh")
```

```
statechoosen.current()
```

```
statechoosen.place(x=85, y=241)
```

```
Label(registration_screen, text=" ", textvariable=message).place(x= 95, y = 260)
```

```
Button(registration_screen, text="Register", width=10, height=1, bg = "orange",  
command=registration).place(x = 110, y =301 )
```

```
registration_screen.mainloop()
```

```
registrationform()
```

Output:

A screenshot of a Python GUI application titled 'Registration form'. The window has a title bar with a pencil icon and standard window controls. Below the title bar is a yellow header with the text 'Please Enter the details below'. The form contains several input fields: 'Name' with the value 'Rudradev Arya', 'contact' with '8390712030', 'email' with 'rudradev93@gmail.com', 'Gender' with radio buttons for 'Male' (selected) and 'Female', 'City' with a dropdown menu showing 'Pune', and 'State' with a dropdown menu showing 'Maharashtra'. Below these fields is a message 'Data Stored Successfully' and a yellow 'Register' button.

Conclusion: We have successfully learnt how to create Python GUI program for the student registration and store the details in the file thereby Allowing the users to read the details from the file.